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VisConf: Quality-Aware Consensus for Best-of-N in Vision-Language Model Reasoning

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Abstract

Best-of-N methods improve text-only LLM reasoning by sampling multiple trajectories before selecting an answer. For Vision-Language Models (VLMs), answer frequency and model confidence alone can be insufficient, as sampled generations also vary in their engagement with the visual input. We find that distributional confidence and visual attention are complementary indicators of rollout quality. Each is limited on its own, while trajectories that show strong signals in both are more likely to be correct. Motivated by this, we introduce VisConf, a rollout-quality score that multiplicatively combines visual engagement with candidate-calibrated Self-Certainty, penalizing rollouts that are weak in either signal. We incorporate Rank-Weighted Consensus (RWC), a tuning-free aggregation rule that converts within-question VisConf ranks into answer-level vote weights. Across three VLMs on MathVista, MMMU-Pro, and MMStar, VisConf with RWC consistently outperforms strong BoN baselines. Our analysis shows particularly large gains when correct trajectories are present but underrepresented in the candidate pool. Code.

1 Introduction

A widely used test-time scaling method in Large Language Model (LLM) reasoning is Best-of-N (BoN) selection, where multiple candidate trajectories are sampled for each query [Snell et al., 2024, Brown et al., 2024]. Generating multiple rollouts using stochastic decoding provides more opportunities to produce correct reasoning paths, but also introduces a selection problem since repeated sampling can produce redundant reasoning patterns rather than genuinely diverse solutions [Bilal et al., 2026]. Vision-Language Models (VLMs) similarly benefit from rollout scaling [Sammani et al., 2026, Tong et al., 2026], but multimodal reasoning introduces another dimension of candidate variation as trajectories can differ in how strongly they engage with the visual input.

Two natural signals for assessing VLM rollout quality are confidence and visual engagement. Output-probability-based confidence captures how decisively the model supports its generated trajectory, and has been widely used as a proxy for reliability [Geng et al., 2024, Kang et al., 2025]. However, high confidence does not necessarily imply correct visual reasoning [Cho et al., 2026]. Visual attention can indicate greater reliance on the image, but may also reflect repeated inspection caused by uncertainty or misinterpretation. Our empirical analysis shows that these signals are weakly informative but provide complementary information at the trajectory level: rollouts exhibiting both high confidence and strong visual engagement are substantially more reliable than those strong in only one dimension. We further find that their useful interaction is primarily trajectory-level rather than token-aligned. Motivated by these findings, we introduce VisConf, which independently aggregates confidence and visual engagement over a rollout before combining them into a single quality score.

For verifiable tasks, BoN exposes two sources of evidence: the quality of individual rollouts and agreement across the candidate set. Self-Consistency exploits the latter by favoring answers supported

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by multiple independently sampled trajectories, under the intuition that diverse reasoning paths converging to the same answer provide stronger evidence of correctness [Wang et al., 2023]. However, standard Self-Consistency treats all rollouts equally and fails when redundant incorrect generations form a majority group [Yin et al., 2024, Taubenfeld et al., 2025]. In contrast, conventional BoN selection relies only on the single highest-scoring rollout, discarding corroborating evidence from other trajectories that share the same outcome. We therefore use Rank-Weighted Consensus (RWC), a Borda-inspired hyperparameter-free aggregation scheme, to combine trajectory quality with answer-level consensus.

Our contributions are as follows: (1) We empirically characterize confidence and visual engagement as complementary trajectory-level signals of VLM rollout quality, showing that their useful interaction is primarily trajectory-level rather than token-aligned. (2) We introduce a VLM BoN framework featuring VisConf, which combines candidate-calibrated Self-Certainty with visual engagement, and RWC, which incorporates trajectory quality into answer-level consensus. (3) We demonstrate consistent improvements across three models, three benchmarks, and three sampling budgets, including average accuracy gains of 0.98 points over Self-Consistency and a statistically significant 0.49 points over the strongest baseline.

2 Related Work

Test-Time Scaling and Best-of-N Test-time scaling improves performance by allocating additional inference compute without modifying model parameters [Snell et al., 2024, Brown et al., 2024]. Self-Consistency aggregates candidates through answer frequency, but provides no explicit signal of rollout quality [Wang et al., 2023]. Alternatively, verifier- and reward-model-based methods rank candidates using learned correctness features, at the cost of training or invoking an additional model [Cobbe et al., 2021, Lightman et al., 2023]. Recent work in VLMs finds modest gains from single-model scaling while evaluating lexical/length-based heuristics and majority voting, leaving stronger model-internal rollout-quality signals unexplored [Tong et al., 2026].

Confidence-Based Selection Model confidence provides a verifier-free quality signal for sampled trajectories [Geng et al., 2024, Kang et al., 2025]. Likelihood-based measures such as sequence log-probability use only the probabilities assigned to the sampled tokens, ignoring the remainder of the predictive distribution [Kang et al., 2025, Kim and Kim, 2026]. Self-Certainty instead uses the full predictive distribution through Kullback-Leibler (KL) divergence from a uniform reference distribution [Kang et al., 2025]. However, its Borda-based aggregation introduces a tunable exponent.

Visual Grounding in VLMs VLM responses can be weakly image-conditioned, exhibiting hallucination and language-prior bias, motivating methods that promote visual faithfulness [Li et al., 2023, Leng et al., 2024]. VGPO identifies temporal visual forgetting, showing that sustained visual attention can be indicative of correctness [Wang et al., 2026]. Most closely related, LookBack combines token likelihood with attention-based visual reference to select the highest-scoring rollout [Cho et al., 2026]. In contrast, VisConf calibrates distributional confidence within the candidate set, aggregates confidence and visual engagement separately over each rollout before fusion, and uses RWC to retain answer-level consensus.

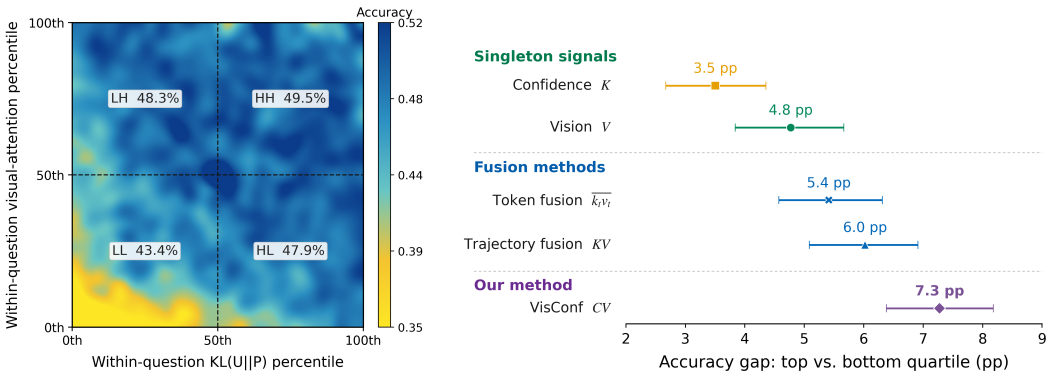
3 What Signals VLM Rollout Quality?

BoN succeeds only when the selection rule can distinguish reliable reasoning from plausible errors. During autoregressive generation, the transformer architecture exposes two distinct sources of information at each decoding step: its predictive distribution over the vocabulary, from which we derive *confidence*, and its internal attention to the visual input, which we term *visual engagement*.

3.1 Standalone Quality Signals

Confidence via Self-Certainty For a generation trajectory i at step t, let $p_{i,t}$ denote the next-token distribution obtained by applying softmax directly to the unscaled model logits, and U the uniform distribution over the vocabulary. The token-level confidence signal is

$$k_{i,t} = D_{\mathrm{KL}}(U \,\|\, p_{i,t})$$



(a) Confidence-vision accuracy landscape

(b) Top-bottom quartile accuracy gap

Figure 1: Analysis on Qwen2.5-VL-3B using MathVista, MMMU-Pro, and MMStar. (a) Accuracy as a function of confidence and visual engagement. (b) Discriminative strength measured by the accuracy gap between the top and bottom score quartiles. Combining confidence and visual engagement is more informative, but how the signals are combined also matters.

where larger values indicate more concentrated predictive distributions [Kang et al., 2025].

Visual Engagement via Visual Attention We use visual attention as a token-level measure of visual engagement, following LookBack [Cho et al., 2026]. Let $\alpha_{i,t,j}^{(\ell,h)}$ be the post-softmax attention from the position predicting token t to context position j in layer ℓ and head h. We average attention across all layers and heads and sum within each token group S:

$$\bar{\alpha}_{i,t,j} = \frac{1}{\sum_\ell H_\ell} \sum_\ell \sum_{h=1}^{H_\ell} \alpha_{i,t,j}^{(\ell,h)}, \qquad A_{i,t}^{S} = \sum_{j \in S} \bar{\alpha}_{i,t,j}.$$

For image ($\mathcal{I}$), prompt-text ($\mathcal{P}$), and all previously generated-token ($\mathcal{G}_t$) positions, excluding padding and non-image special tokens, we define token-level visual engagement as

$$v_{i,t} = \frac{A_{i,t}^{\mathcal{I}}}{A_{i,t}^{\mathcal{I}} + A_{i,t}^{\mathcal{P}} + A_{i,t}^{\mathcal{G}_t}}.$$

Larger $v_{i,t}$ indicates a greater share of attention allocated to the image relative to textual context.

We then average the token-level signals for a generation i to obtain raw confidence and visual engagement scores:

$$K_i = \frac{1}{T_i} \sum_{t=1}^{T_i} k_{i,t}, \qquad V_i = \frac{1}{T_i} \sum_{t=1}^{T_i} v_{i,t}.$$

Top-1 BoN selection chooses rollout $\hat{i}$ among N rollouts using $\hat{i} = \arg\max_{i \in [1,N]} Q_i$, where $Q_i = K_i$ for confidence-based selection and $Q_i = V_i$ for visual engagement-based selection.

3.2 Each Signal is Informative But Weak

We first examine how strongly confidence and visual engagement individually predict correctness. We measure how well trajectory-level scores separate correct from incorrect rollouts using the accuracy gap between the top and bottom score quartiles. A useful quality signal should concentrate more correct rollouts in its highest-scoring quartile. We additionally report macro-averaged prompt-level AUC in the appendix (Table 6).

Mean confidence, K_i, produces an average top-bottom accuracy gap of 3.5 percentage points (Figure 1b). High confidence may reflect a coherent and internally consistent reasoning path, but VLMs can also be confidently wrong when language priors dominate or an early visual misinterpretation propagates through otherwise fluent reasoning. Mean visual engagement, V_i, yields a slightly larger gap of 4.8 points. Sustained image attention may indicate continued use of visual evidence, but can

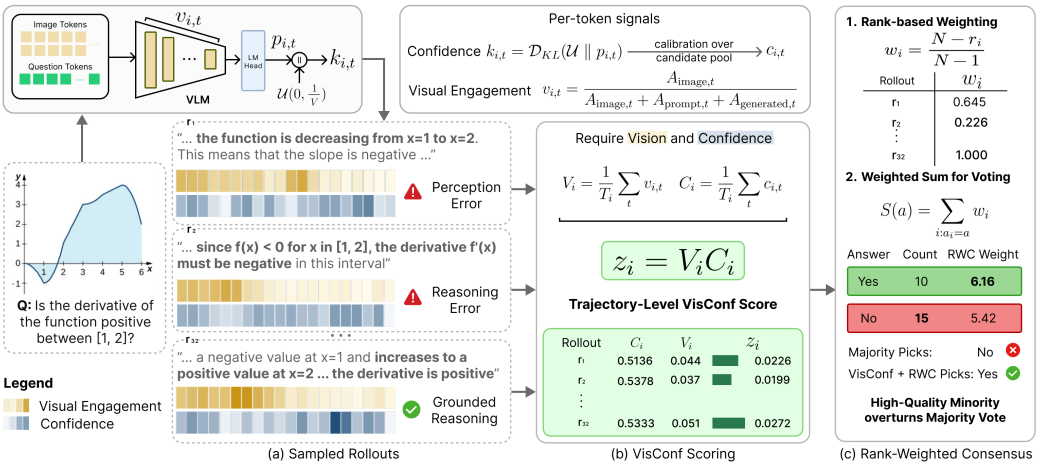


Figure 2: Overview of the VisConf and RWC framework. Given multiple sampled VLM rollouts, VisConf aggregates calibrated confidence and visual engagement independently over each trajectory, and combines them into a rollout-quality score, rewarding trajectories that are both confident and visually engaged. RWC uses VisConf scores to effectively select high-quality answer groups.

also reflect uncertainty, repeated reinspection, or attention to irrelevant content. Conversely, some correct rollouts may also extract necessary visual information during early reasoning and therefore not require prolonged visual engagement.

3.3 Confidence and Vision are Complementary at the Trajectory Level

Joint Signal Structure The two signals become notably informative when considered jointly. We stratify rollouts by whether their within-prompt raw-confidence and visual engagement scores fall above or below the median. Rollouts high in both signals achieve 49.5% accuracy, compared with 48.3% when only visual engagement is high, 47.9% when only confidence is high, and 43.4% when both are low (Figure 1a). Holding either signal fixed, higher values of the other are also associated with greater correctness enrichment (Appendix Figure 6). Together, these results indicate that confidence and visual engagement provide complementary rather than redundant evidence of rollout quality.

Trajectory-Level vs. Token-Level Fusion The joint analysis motivates combining confidence and visual engagement, but does not specify where fusion should occur. Token-level fusion averages the stepwise products, $Q_i^{\text{tok}} = \frac{1}{T_i}\sum_t v_{i,t}k_{i,t}$, coupling the two signals at the same decoding steps. Trajectory-level fusion instead multiplies their rollout means, $Q_i^{\text{traj}} = V_i K_i$, allowing confidence and visual engagement to contribute at different stages of generation. Both improve separation over either singleton signal, but trajectory-level fusion produces a larger top–bottom quartile accuracy gap than token-level fusion (6.0 vs. 5.4 percentage points; Figure 1b).

The two constructions differ only through within-rollout temporal covariance: trajectory-level fusion retains the marginal strength of each signal without requiring their tokenwise peaks to coincide. Consistent with this decomposition, randomizing token correspondence matches the AUC of trajectory-level fusion, whereas aligned token fusion is slightly weaker (Appendix D.2). This suggests that the useful interaction between confidence and visual engagement lies primarily in their trajectory-level magnitudes.

4 VisConf: Trajectory-Level Visual-Confidence Scoring

Shaped by our analysis in Section 3, we introduce VisConf, which first calibrates confidence relative to the candidate set and then combines rollout-level confidence and visual engagement (Figure 2).

4.1 Candidate-Relative Confidence Calibration

To integrate Self-Certainty into a multi-signal score, we calibrate confidence relative to the pooled token-level values across all candidate rollouts G for a query.

$$c_{i,t} = \text{clip}_{[0,1]} \left(\frac{k_{i,t} - Q_{.05}(G)}{\max\left(Q_{.95}(G) - Q_{.05}(G),\, \epsilon\right)} \right),$$

where $Q_p(G)$ is the p-quantile of pooled token KL values in G, and $\epsilon > 0$ ensures numerical stability. This scales confidence to $[0, 1]$ while bounding the influence of extreme KL values. Using shared percentiles across G therefore situates each rollout in the context of the candidate set, making subsequent trajectory-level scores relative rather than absolute.

4.2 Trajectory-Level Visual-Confidence Fusion

VisConf aggregates visual engagement and calibrated confidence independently over a complete rollout by taking the average across the full generation, resulting in V_i and C_i, respectively. We then define the VisConf quality score as:

$$Z_i = V_i C_i.$$

VisConf favors trajectories that exhibit both high calibrated confidence and high visual engagement, without explicitly requiring that the two signals coincide at the same decoding steps.

5 From Rollout Quality to Answer Consensus

VisConf estimates the relative quality of each rollout, but selecting only the highest-scoring candidate treats this noisy score as a verifier and ignores corroborating evidence from other trajectories. The value of retaining both sources of evidence is reflected in their joint association with rollout correctness (Appendix Figure 8). Inspired by Borda voting [Kang et al., 2025], we present *Rank-Weighted Consensus* (RWC) to consolidate consensus with quality, as illustrated in Figure 2(c).

5.1 Rank-Based Quality Weighting

Let r_i denote the rank of a quality score Z_i among N rollouts, with $r_i = 1$ being the highest-ranked candidate. We convert each rank into a normalized quality weight:

$$w_i = \frac{N - r_i}{N - 1},$$

with exact score ties receiving their average rank. The highest-ranked rollout receives weight 1 and the lowest receives 0. Lower-ranked rollouts contribute proportionally less. Because these weights depend only on ordering, they are invariant to strictly increasing transformations of the VisConf score. Unlike Self-Certainty's parameterized Borda weights $b_p(r_i) = (N - r_i + 1)^p$, RWC uses a fixed linear schedule that downweights weak evidence without requiring p to be tuned [Kang et al., 2025].

5.2 Quality-Weighted Answer Consensus

Let a_i denote the normalized answer extracted from rollout i. We sum the quality weights of all trajectories supporting each answer and select the answer with the highest quality-informed support:

$$\text{Vote}(a) = \sum_{i=1}^{N} \mathbf{1}[a_i = a]\, w_i, \qquad \widehat{a} = \arg\max_a \text{Vote}(a).$$

Thus, repeated answers accumulate support, but higher-quality trajectories exert more influence on the final prediction. A minority answer can therefore prevail when its quality compensates for its lower frequency.

6 Experimental Setup

We evaluate response-selection methods in VLMs under a shared evaluation protocol.

Table 1: Averaged accuracy (%) across MathVista, MMMU-Pro, and MMStar at varying rollout budgets N. Per-dataset results are presented in Appendix B.1. VisConf + RWC achieves the highest average for every model and sampling budget.

Method	**Qwen2.5-VL-3B**			**InternVL3-8B**			**Gemma4-E2B**			**Avg.**
	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	
FirstAns	48.51	48.51	48.51	56.55	56.55	56.55	47.34	47.34	47.34	50.80
Self-Consistency	53.70	54.75	55.07	60.96	61.71	61.80	49.85	50.23	50.35	55.38
Self-Certainty	49.78	49.95	49.94	58.83	59.03	59.07	49.43	49.62	49.81	52.83
+ Borda ($p=0.3$)	53.83	54.87	55.14	61.21	61.97	62.25	50.12	50.50	50.71	55.62
+ Borda ($p=1.2$)	53.53	54.72	55.33	61.21	62.03	62.78	50.35	50.82	50.91	55.74
LookBack	50.09	50.34	50.57	58.22	58.11	57.84	49.16	49.74	50.27	52.70
+ RWC	53.68	54.93	55.72	61.27	62.16	62.65	50.30	50.86	51.19	55.86
VisConf	51.04	51.16	51.17	59.46	59.81	60.27	50.07	50.49	50.58	53.78
+ RWC	**54.09**	**55.32**	**56.29**	**61.69**	**62.53**	**63.27**	**50.89**	**51.44**	**51.68**	**56.36**

Models and Benchmarks: We evaluate Qwen2.5-VL-3B [Bai et al., 2025], InternVL3-8B [Zhu et al., 2025], and Gemma4-E2B [Gemma Team et al., 2026] on three multimodal reasoning benchmarks: MathVista [Lu et al., 2024], MMMU-Pro [Yue et al., 2025], and MMStar [Chen et al., 2024], using the MathVista `testmini` split and the full evaluation sets of MMMU-Pro and MMStar.

Candidate Generation and Evaluation: For each question, we generate 32 chain-of-thought rollouts [Wei et al., 2022, Zhang et al., 2024] using a fixed stochastic decoding configuration across models. We use temperature = 0.7, top-$p = 0.8$, top-$k = 20$, and a maximum of 2048 generated tokens. We report selection accuracy for $N \in \{8, 16, 32\}$. For $N = 8$ and $N = 16$, results are averaged over 100 fixed subsets sampled without replacement from the 32-rollout pool using fixed seeds. $N = 32$ uses the complete pool once. Accuracy is averaged within each question before dataset-level aggregation.

Baselines: We compare against *FirstAns*, which is the expected accuracy of one randomly sampled rollout, and *Self-Consistency*, which selects the most frequent answer. We evaluate *Self-Certainty* using top-1 BoN selection as well as Borda-based aggregation with $p \in \{0.3, 0.7, 1.2, 2.0\}$[1] [Kang et al., 2025]. For multimodal scoring, we evaluate *LookBack* [Cho et al., 2026] using its original top-1 selection and additionally pair its scores with RWC, allowing comparison under the same answer-aggregation rule. Finally, we evaluate VisConf with top-1 selection and RWC, recomputing confidence normalization and ranks within each sampled candidate subset.

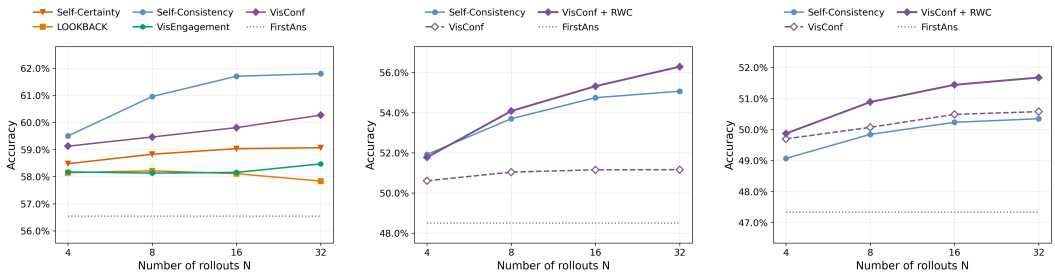
7 Results

7.1 Overall Performance and Scaling

Table 1 reports accuracy averaged across the three benchmarks. VisConf + RWC achieves the highest accuracy for every model and sampling budget, with an overall average of 56.36%. This exceeds LookBack + RWC, the strongest baseline overall, by 0.5 points; the best Self-Certainty + Borda configuration by 0.62 points; and Self-Consistency by 0.98 points. Across the per-dataset results (Appendix Table 4), VisConf + RWC outperforms every baseline in 24 of 27 model-dataset-budget settings and trails by at most 0.27 points in the remaining three. Improvement against the strongest baseline remains statistically significant (paired 95% CI: $[0.32,\ 0.66]$) and positive at every sampling budget (Appendix B.2).

Comparing top-1 selection without aggregation, VisConf achieves an overall average of 53.78%, 1.08 and 0.95 points higher than LookBack (52.70%) and Self-Certainty (52.83%), respectively. VisConf's advantage over LookBack suggests that independently high confidence and visual engagement across decoding stages yields a stronger quality signal than requiring both signals to coincide token-wise. These top-1 results encourage the potential use of VisConf where aggregation schemes are not applicable. In our settings, however, omitting answer-group information is generally restrictive, as

[1] Results for $p = 0.7$ and $p = 2.0$ are reported in Appendix B.1.



(a) Top-1 Selection: InternVL3-8B (b) Aggregated: Qwen2.5-VL-3B (c) Aggregated: Gemma4-E2B

Figure 3: Effect of scaling N on accuracy across three benchmarks. (a) VisConf top-1 selection scales strongly but underperforms Self-Consistency. (b-c) VisConf + RWC consistently outperforms top-1 and Self-Consistency, with its advantage persisting or increasing with scaling.

top-1 VisConf selection performs below VisConf + RWC across all 27 settings. This validates the use of quality-informed aggregation in tasks where outcome-level rollout grouping is available.

VisConf + RWC provides additional benefits as N is scaled. From $N = 8$ to $N = 32$, VisConf + RWC improves from 54.09% to 56.29% on Qwen2.5-VL-3B, 61.69% to 63.27% on InternVL3-8B, and 50.89% to 51.68% on Gemma4-E2B (Figure 3). Our formulation at $N = 16$ outperforms Self-Consistency at $N = 32$ in all three models, indicating more effective use of the rollout budget.

7.2 Ablation Study

Score Ablation Table 2 shows the results of isolating each VisConf score component under top-1 selection on Qwen2.5-VL-3B. Visual engagement and confidence without fusion are relatively weak selectors, at 49.58% and 49.89%, while their raw trajectory-level product improves performance to 50.12%. Incorporating candidate-relative calibration with token-level fusion achieves 50.44%, lower than our VisConf trajectory-level formulation, which is 0.68 points higher at 51.12%. These results validate individual components of VisConf as meaningfully contributing to the overall performance. Our VisConf score can be generally written as

$$Z_{i,\alpha} = V_i^{\alpha} C_i^{1-\alpha},$$

where α controls the relative influence of visual engagement and confidence. Since $Z_{i,0.5} = \sqrt{V_i C_i}$, it induces exactly the same ranking as the product $V_i C_i$ used by VisConf. A post-hoc sensitivity sweep validates this balanced choice without per-dataset tuning (Appendix Figure 7).
Furthermore, to assess whether VisConf's performance can be attributed to generation length bias, we include a MinLength baseline that selects the shortest rollout. Its substantially lower accuracy (49.20%) suggests that VisConf's gains are not driven by a simple preference for shorter generations.

Aggregation Ablation Holding the quality score fixed and replacing RWC with raw-score weighted consensus WC (where $w_i = Z_i$) reduces accuracy from 55.23% to 55.02%: ranks preserve the candidate ordering while discarding unreliable score magnitudes.

Table 2: Ablation study on Qwen2.5-VL-3B. Average accuracy % across the three benchmarks for $N \in \{8, 16, 32\}$.

Method	Avg.
Score Ablation with BoN	
Visual Engagement: V	49.58
Self-Certainty: K	49.89
VisConf w/o KL Calibration: VK	50.12
Token Fusion: $\overline{v_{i,t} c_{i,t}}$	50.44
VisConf: VC	**51.12**
MinLength	49.20
Aggregation Ablation	
VisConf + WC	55.02
VisConf + RWC	**55.23**

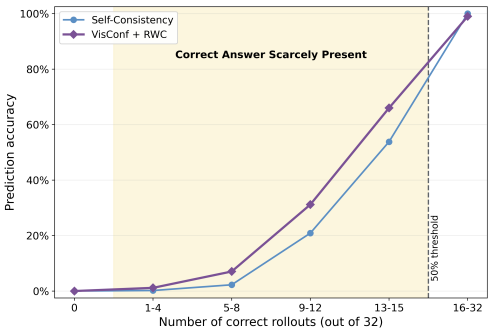


Figure 4: Performance comparison against Self-Consistency on InternVL3-8B in cases where the candidate pool contains few correct rollouts. VisConf + RWC effectively prioritizes correct evidence without majority support.

7.3 Performance When Correct Evidence Is Sparse

Figure 4 examines how performance varies with empirical generation difficulty, measured by the prevalence of correct-answer trajectories in the $N = 32$ candidate pool,

$$g_q = \frac{n_{\text{correct}}}{N}.$$

VisConf + RWC increasingly outperforms Self-Consistency as correct evidence approaches, but remains below, the absolute majority threshold. Its advantage rises from 0.96 percentage points with 1-4 correct rollouts to 4.79 points with 5-8, 10.29 points with 9-12, and 12.18 points with 13-15. All 110 questions rescued relative to Self-Consistency occur within $0 < g_q < 0.5$, compared with 29 regressions, yielding 81 net corrections in this subset (Appendix Table 7). Thus, VisConf + RWC provides its largest gains when substantial correct evidence exists but remains outnumbered under unweighted consensus, using quality information to prioritize stronger evidence over frequency alone.

8 Discussion

Best-of-N is both a discovery and an aggregation problem. Increasing the rollout budget results in two competing outcomes: it increases the chance of discovering a correct reasoning path, but is likely to introduce additional incorrect candidates which may also be high-scoring. Our direct-selection results demonstrate that simply extending the candidate set does not guarantee better BoN selection, even when the oracle accuracy of the pool continues to increase (Appendix B.1). The stronger scaling of VisConf + RWC suggests that the value of additional test-time compute depends not only on generating better candidates, but also on preserving useful evidence distributed across the candidate set. We observe that this preservation can positively affect outcomes in cases where correct reasoning is less commonly encountered.

Relative quality is more reliable than absolute quality. Confidence and visual engagement are noisy proxies, and neither of them provides a reliable estimate that a rollout is correct. VisConf and RWC therefore use these signals comparatively while carefully avoiding spurious conclusions. Candidate-relative calibration contextualizes confidence within the current pool, multiplicative fusion promotes the occurrence of both signals, and RWC retains only ordinal rankings rather than interpreting raw magnitudes. This separation of roles is important. VisConf provides a general measure of which trajectories appear stronger, while RWC gauges how this imperfect ordering aligns with answer-level agreement. An extreme raw score's domination is therefore subdued (Appendix E.3) and any single trajectory's outcome is not treated as conclusive. Our work suggests that effective selection is possible without requiring a perfect verifier. Weak model-internal signals can become useful provided that the influence of noise is systematically reduced.

Limitations and Future Work. While we incorporate attention as a reference for the model engaging with input image(s), attention alone does not establish correct visual understanding or faithfully explain the model's reasoning [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019]. Obtaining visual-attention signals also requires eager attention extraction. In a controlled replay study on Qwen2.5-VL-3B, the median runtime overhead relative to FlashAttention-2 was negligible at $N = 8$ and $N = 16$ but increased to 18.4% at $N = 32$ (Appendix C). Future work could replace attention with a cheaper proxy or one that more causally explains whether visual evidence is reliably used. Although VisConf's superior top-1 performance over other scoring baselines suggests that it may remain useful when answer-level grouping is unavailable, we do not evaluate this setting directly. More broadly, VisConf provides a training-free estimate of rollout quality that need not be confined to selection, and future work could study its use as a reward signal for policy optimization or in other paradigms where judging trajectory quality is useful.

9 Conclusion

We introduced VisConf, a simple, training-free response quality score using model-internal quality signals. Together with RWC, our framework combines rollout quality with answer-level agreement to improve BoN selection while requiring no tunable hyperparameters or external reward models. Our results demonstrate that imperfect model-internal signals can support effective test-time scaling when used comparatively and accumulated across agreeing trajectories rather than treated as standalone verifiers.

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A Experimental and Implementation Details

Prompt templates Each prompt contains the image(s), question, and answer choices when applicable. Qwen2.5-VL and InternVL3 receive the following chain-of-thought prompt: "Think step by step inside <think> </think> tags, then give your final answer inside \boxed{}." Gemma4 is run with thinking mode disabled and receives the same instruction without explicit thinking tags: "Think step by step, then give your final answer inside \boxed{}." Inputs are rendered using each model's native chat template.

Dataset information We evaluate the 1,000-example MathVista `testmini` split, the complete 1,730-example MMMU-Pro standard test set, and the complete 1,500-example MMStar validation set for all three models.

MathVista evaluates mathematical reasoning in visual contexts and contains both multiple-choice and free-form questions. It contains arithmetic, algebraic, geometric, logical, numerical, statistical, and scientific reasoning problems, with visual inputs including diagrams, charts, tables, function plots, scientific figures, natural images, and puzzle-style scenes [Lu et al., 2024].

MMMU-Pro targets college-level reasoning across a broad range of disciplines, including science, engineering, medicine, business, humanities, and the social sciences. Questions require combining domain knowledge with information from various visual inputs such as diagrams, plots, tables, maps, and scientific figures. The benchmark is constructed from MMMU by filtering questions that can be answered reliably without the image, making visual evidence more necessary for solving the task [Yue et al., 2025].

MMStar consists of visually dependent multiple-choice questions organized across six capability categories: coarse perception, fine-grained perception, instance reasoning, logical reasoning, science and technology, and mathematics. These are further divided into 18 fine-grained dimensions covering tasks such as object localization and counting, attribute recognition, cross-instance comparison, diagram reasoning, numerical calculation, and statistical analysis. Its examples are curated to reduce language-only shortcuts and ensure that visual information is necessary for answering the question [Chen et al., 2024].

Answer extraction and normalization We extract the last well-formed $\texttt{\textbackslash boxed}\{\cdot\}$ expression in each response. Multiple-choice answers are canonicalized to uppercase option letters. Boxed option text is matched to its corresponding choice. When no valid boxed output is present, conservative final-sentence and final-line patterns are used. Free-form MathVista answers are compared using normalized numerical or string matching, including fraction handling and decimal comparison. Unextractable responses are assigned a shared `<NO_ANSWER>` label.

Tie handling Direct Best-of-N selection resolves score ties using the lowest original rollout index. For RWC, equal VisConf scores receive their average rank; if all scores are equal, every rollout receives weight 0.5. If answer-level votes are tied, we select the answer whose earliest supporting rollout has the lowest original index.

Compute Resources Experiments with Qwen2.5-VL-3B and Gemma4-E2B used one NVIDIA GeForce RTX 4090 GPU with 24 GB of memory; InternVL3-8B experiments used one NVIDIA A100 GPU with 80 GB of memory.

Table 3: LookBack Best-of-N and Rank-Weighted Consensus (RWC) performance under two values of α, with $\lambda = 1.25$.

α	λ	Aggregation	Qwen2.5-VL-3B			InternVL3-8B			Gemma4-E2B			Avg.
			$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	
0.25	1.25	BoN	49.88	50.17	50.45	56.31	55.00	52.99	48.92	49.38	49.95	51.45
0.25	1.25	RWC	53.40	54.61	55.51	60.78	61.68	62.07	50.16	50.70	50.84	55.53
0.50	1.25	BoN	50.09	50.34	50.57	58.22	58.11	57.84	49.16	49.74	50.27	52.70
0.50	1.25	RWC	53.68	54.93	55.72	61.27	62.16	62.65	50.30	50.86	51.19	55.86

LookBack Implementation For LookBack, let $\ell_{i,t}$ be the sampled-token log-probability and $A_{i,t}$ the all-layer, all-head image-attention mass. Its response score is

$$L_i = \frac{\sum_t A_{i,t}^{\lambda} \left(\ell_{i,t} + \alpha \log A_{i,t}\right)}{\sum_t A_{i,t}^{\lambda}}.$$

The work suggests a choice of $\alpha = 0.25$ and $\lambda = 1.25$ as a global setting. However, we observe that $\alpha = 0.5$ and $\lambda = 1.25$, which were the model-specific settings for Qwen2.5-VL-7B, achieve greater performance (Table 3). Thus, we fix $\alpha = 0.5$ and $\lambda = 1.25$ across all models, datasets, and sampling budgets in our baseline results. Standard LookBack selects the highest-scoring rollout. LookBack + RWC applies our rank-weighted answer aggregation to the same L_i scores.

B Full Results and Statistical Analysis

B.1 Per-Model, Dataset, and N Results

Table 4: Full accuracy results (%) on MathVista, MMMU-Pro, and MMStar for $N \in \{8, 16, 32\}$. Avg. is the unweighted mean across the nine dataset-budget combinations within each model.

Method	MathVista			MMMU-Pro			MMStar			Avg.
	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	
Qwen2.5-VL-3B										
Oracle	82.76	87.01	89.40	80.31	88.10	92.95	85.50	91.57	95.67	88.14
FirstAns	54.01	54.01	54.01	41.20	41.20	41.20	50.31	50.31	50.31	48.51
Self-Consistency	61.04	62.48	63.20	45.30	46.38	46.88	54.77	55.38	55.13	54.51
Self-Certainty	55.65	55.70	55.40	42.97	43.25	43.76	50.72	50.89	50.67	49.89
+ Borda ($p=0.3$)	**61.35**	62.78	63.60	45.52	46.55	46.76	54.63	55.27	55.07	54.61
+ Borda ($p=0.7$)	61.25	62.79	63.90	45.60	46.64	47.28	54.48	55.08	54.93	54.66
+ Borda ($p=1.2$)	60.88	62.61	63.90	45.55	46.58	47.23	54.15	54.96	54.87	54.53
+ Borda ($p=2.0$)	60.07	62.26	63.90	45.28	46.42	47.23	53.68	54.72	55.07	54.29
LookBack	55.56	55.64	55.50	42.98	43.22	43.41	51.74	52.17	52.80	50.33
+ RWC	60.66	62.44	63.70	45.75	46.90	47.34	54.64	55.44	**56.13**	54.78
VisConf	56.63	56.82	57.10	43.57	43.44	43.06	52.93	53.21	53.33	51.12
+ RWC	61.08	**62.97**	**64.50**	**46.12**	**47.24**	**48.38**	**55.06**	**55.75**	56.00	**55.23**
InternVL3-8B										
Oracle	80.46	83.92	86.50	80.69	86.67	90.64	84.77	88.81	91.80	86.03
FirstAns	57.57	57.57	57.57	49.43	49.43	49.43	62.65	62.65	62.65	56.55
Self-Consistency	63.62	64.66	64.60	53.28	54.02	54.34	65.97	66.44	66.47	61.49
Self-Certainty	60.75	60.79	60.50	51.68	52.15	52.37	64.05	64.16	64.33	58.98
+ Borda ($p=0.3$)	64.05	65.12	65.70	53.55	54.24	54.45	66.03	66.55	66.60	61.81
+ Borda ($p=0.7$)	64.15	65.25	66.10	53.66	54.38	54.62	66.06	66.57	66.73	61.95
+ Borda ($p=1.2$)	63.99	65.19	66.40	53.62	54.43	55.14	66.03	66.49	66.80	62.01
+ Borda ($p=2.0$)	63.58	64.91	66.20	53.41	54.33	55.20	65.83	66.45	66.60	61.84
LookBack	59.46	59.34	59.80	50.85	50.82	50.17	64.34	64.17	63.53	58.05
+ RWC	63.71	65.00	65.90	53.72	54.60	54.97	66.38	66.88	67.07	62.02
VisConf	61.04	61.26	61.40	52.25	52.90	53.41	65.10	65.27	66.00	59.85
+ RWC	**64.29**	**65.49**	**67.00**	**54.13**	**54.96**	**55.55**	**66.65**	**67.14**	**67.27**	**62.50**
Gemma4-E2B										
Oracle	74.64	80.08	83.50	72.81	79.87	84.97	79.38	84.96	88.73	80.99
FirstAns	47.63	47.63	47.63	41.90	41.90	41.90	52.49	52.49	52.49	47.34
Self-Consistency	50.14	50.56	50.90	44.29	44.59	44.68	55.10	55.55	55.47	50.14
Self-Certainty	49.21	49.43	50.60	43.77	43.87	43.76	55.30	55.55	55.07	49.62
+ Borda ($p=0.3$)	50.34	50.76	50.90	44.49	44.83	45.38	55.54	55.90	55.87	50.44
+ Borda ($p=0.7$)	50.44	50.89	50.80	44.62	45.03	45.49	55.75	56.18	56.13	50.59
+ Borda ($p=1.2$)	50.49	50.90	50.60	44.69	45.16	45.32	55.87	56.38	56.80	50.69
+ Borda ($p=2.0$)	50.44	50.87	50.80	44.71	45.23	45.78	55.94	56.54	**56.87**	50.80
LookBack	48.98	49.40	49.80	44.00	44.97	46.01	54.48	54.85	55.00	49.72
+ RWC	50.35	50.76	51.10	45.00	45.66	46.07	55.54	56.15	56.40	50.78
VisConf	49.95	50.54	50.80	44.87	45.25	45.20	55.39	55.68	55.73	50.38
+ RWC	**50.93**	**51.48**	**51.70**	**45.64**	**46.22**	**46.53**	**56.09**	**56.63**	56.80	**51.34**

B.2 Paired Statistical Analysis

We estimate uncertainty using a paired question-cluster bootstrap (Table 5). For $N \in \{8, 16\}$, accuracy is first averaged over the 100 shared candidate subsets for each question; $N = 32$ uses the complete rollout pool. We resample the 4,230 questions with replacement within each benchmark,

Table 5: Paired accuracy improvements of VisConf + RWC over each comparator in percentage points with 95% question-cluster bootstrap confidence intervals. Overall averages across sampling budgets, models, and benchmarks.

Comparator	$N = 8$	$N = 16$	$N = 32$	Overall
LookBack + RWC	+0.47 [0.35, 0.59]	+0.45 [0.31, 0.61]	+0.56 [0.24, 0.87]	+0.49 [0.32, 0.66]
Self-Consistency	+0.72 [0.56, 0.86]	+0.87 [0.68, 1.07]	+1.34 [0.97, 1.72]	+0.98 [0.76, 1.18]
Self-Cert. + Borda ($p = 1.2$)	+0.52 [0.40, 0.66]	+0.58 [0.40, 0.74]	+0.74 [0.39, 1.09]	+0.61 [0.43, 0.80]
VisConf (top-1)	+2.03 [1.79, 2.28]	+2.61 [2.21, 3.02]	+3.08 [2.42, 3.77]	+2.57 [2.15, 3.00]

keeping evaluations of the same question paired across models. Method differences are computed before aggregation and macro-averaged across the nine model-benchmark combinations. The overall estimate additionally averages across $N \in \{8, 16, 32\}$. We report percentile 95% confidence intervals from 2,000 bootstrap replicates.

C Computational Overhead of Attention Extraction

We measure the overhead of replacing FlashAttention-2 with eager attention and explicit attention extraction using a controlled replay experiment on Qwen2.5-VL-3B. We replay the exact stored output-token sequences for nested candidate pools of $N \in \{8, 16, 32\}$ on 120 MMMU-Pro prompts, with three repetitions per condition, while holding prompts, images, generated tokens, batching, and KV caching fixed. Figure 5 reports the paired wall-clock runtime ratio of eager attention extraction to FlashAttention-2, computed per prompt after taking the median over repetitions. Median runtime is comparable at $N = 8$ and $N = 16$, with ratios of $0.99\times$ and $1.01\times$, respectively, while the overhead becomes more pronounced at $N = 32$, where eager attention extraction requires $1.18\times$ the runtime. Because the benchmark excludes scoring and answer aggregation, these measurements represent the inference-time cost of obtaining the attention signal itself.

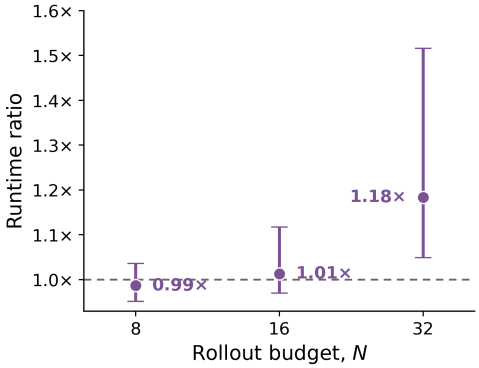


Figure 5: Paired wall-clock runtime ratio of eager attention extraction to FlashAttention-2 on Qwen2.5-VL-3B across 120 MMMU-Pro prompts. Points show median ratios and error bars show the interquartile range.

D Analysis of the VisConf Score

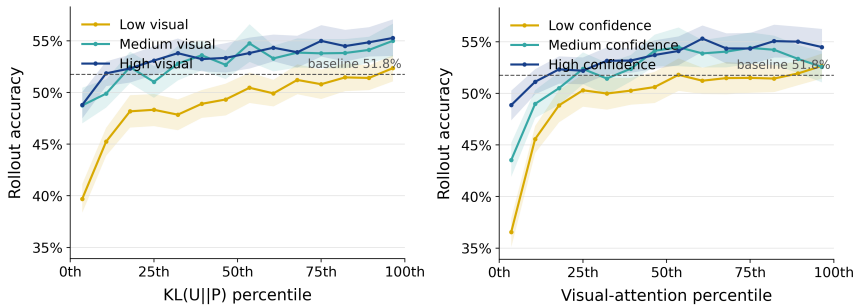
D.1 Joint Enrichment of Confidence and Visual Engagement

Figure 6 examines each signal while stratifying by the other. Rollout accuracy generally increases with both confidence and visual engagement, while weak values of either signal limit accuracy. This complementary enrichment motivates combining the two signals in VisConf.

D.2 Token- and Trajectory-Level Fusion

We analyze the raw confidence signal $k_{i,t}$, before applying the candidate-relative calibration used by VisConf, to isolate the effect of where confidence and visual engagement are fused. For rollout i with T_i generated tokens, define

$$V_i = \frac{1}{T_i} \sum_{t=1}^{T_i} v_{i,t}, \qquad K_i = \frac{1}{T_i} \sum_{t=1}^{T_i} k_{i,t}. \tag{1}$$



(a) Accuracy across confidence percentiles, stratified by visual engagement tercile.

(b) Accuracy across visual engagement percentiles, stratified by confidence tercile.

Figure 6: Joint enrichment of rollout correctness by raw confidence and visual engagement. The dashed line denotes overall rollout accuracy, and shaded regions show 95% confidence intervals.

Token-level and trajectory-level fusion are respectively

$$Q_i^{\text{tok}} = \frac{1}{T_i} \sum_{t=1}^{T_i} v_{i,t} k_{i,t}, \qquad Q_i^{\text{traj}} = V_i K_i. \tag{2}$$

Covariance decomposition By the definition of within-rollout covariance,

$$Q_i^{\text{tok}} = V_i K_i + \text{Cov}_t(v_{i,t}, k_{i,t}). \tag{3}$$

Token-level fusion therefore augments the product of the two trajectory means with a term determined by their temporal co-occurrence. Trajectory-level fusion retains the marginal strength of each signal without requiring them to be large at the same decoding positions.

Permutation identity Let π_i be a uniformly random permutation of the T_i visual engagement positions. Then

$$\mathbb{E}_{\pi_i}\left[\frac{1}{T_i}\sum_{t=1}^{T_i} v_{i,\pi_i(t)} k_{i,t}\right] = V_i K_i. \tag{4}$$

For every position t, the permuted value $v_{i,\pi_i(t)}$ is uniformly distributed over the visual engagement sequence and therefore has expectation V_i. Consequently,

$$\frac{1}{T_i}\sum_t \mathbb{E}_{\pi_i}[v_{i,\pi_i(t)}]\, k_{i,t} \;= \frac{1}{T_i}\sum_t V_i k_{i,t} = K_i V_i. \tag{5}$$

Thus, trajectory-level fusion is the expected token product after temporal correspondence between confidence and visual engagement has been removed. The same identities hold after replacing raw confidence $k_{i,t}$ with calibrated confidence $c_{i,t}$; VisConf uses the resulting trajectory-level product $V_i C_i$.

Within-question discrimination We evaluate these formulations on the 3,558 mixed-correctness questions from Qwen2.5-VL-3B across MathVista, MMMU-Pro, and MMStar at $N = 32$. A mixed-correctness question contains at least one correct and one incorrect rollout.

For question q, let $\mathcal{C}_q$ and $\mathcal{I}_q$ denote its sets of correct and incorrect rollouts. For rollout score s, we compute

$$\text{AUC}_q(s) = \frac{1}{|\mathcal{C}_q|\,|\mathcal{I}_q|}\sum_{i\in\mathcal{C}_q}\sum_{j\in\mathcal{I}_q}\mathbf{1}[s_i > s_j]. \tag{6}$$

We macro-average this quantity across questions so that questions with more correct–incorrect rollout pairs do not receive greater weight. Confidence intervals are obtained by bootstrap resampling questions. For permuted token fusion, we average over 100 independently sampled within-rollout permutations.

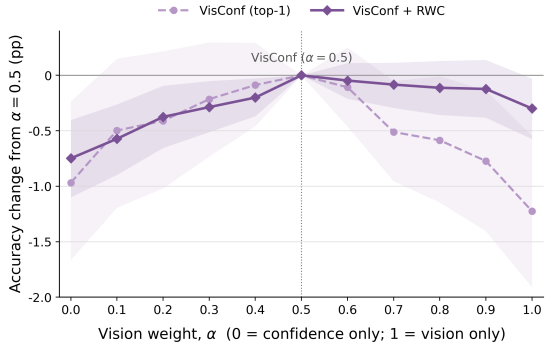


Figure 7: Post-hoc sensitivity of VisConf to the confidence–vision weighting parameter α. Performance is strongest near the balanced setting $\alpha = 0.5$.



Figure 8: Answer correctness by same-answer support and VisConf rank. Accuracy increases along both dimensions.

Table 6: Within-question discrimination of rollout-quality signals on Qwen2.5-VL-3B, pooled across MathVista, MMMU-Pro, and MMStar. Fusion-location controls use raw confidence k, whereas VisConf uses candidate-calibrated confidence c.

Signal	**Score**	**AUC**	**95% CI**
Confidence	K_i	0.5279	[0.5214, 0.5344]
Visual engagement	V_i	0.5307	[0.5243, 0.5374]
Aligned token fusion	$\frac{1}{T_i}\sum_t k_{i,t}v_{i,t}$	0.5373	[0.5304, 0.5440]
Permuted token fusion	$\frac{1}{T_i}\sum_t k_{i,t}v_{i,\pi_i(t)}$	0.5399	[0.5333, 0.5468]
Trajectory fusion	K_iV_i	0.5399	[0.5331, 0.5462]
VisConf	$\mathbf{C_iV_i}$	**0.5503**	**[0.5434, 0.5569]**

Randomizing token correspondence produces essentially the same discrimination as trajectory-level fusion, whereas preserving the original token alignment is slightly weaker. Tokenwise correspondence therefore provides no additional discrimination in our analysis.

The covariance term in Eq. 3 is negatively oriented with respect to correctness. Its within-question AUC is 0.4634, with a 95% confidence interval of $[0.4570, 0.4697]$. The question-macro mean covariance is -0.02659 for correct rollouts and -0.02507 for incorrect rollouts, giving a correct-minus-incorrect difference of -0.00152, with a 95% confidence interval of $[-0.00193, -0.00110]$.

These observations are consistent with the lack of improvement from adding the observed covariance term. They do not imply that correct reasoning requires confidence and visual processing to occur at separate stages. Rather, they show that greater tokenwise co-occurrence may not be a useful quality signal.

D.3 Sensitivity to Confidence-Vision Weighting

To test the sensitivity of VisConf to the relative contribution of confidence and visual engagement, we generalize its multiplicative score as

$$Z_{i,\alpha} = V_i^{\alpha} C_i^{1-\alpha}, \qquad \alpha \in [0, 1].$$

This weighted geometric form preserves the requirement that both signals contribute while allowing their relative importance to vary. The endpoints $\alpha = 0$ and $\alpha = 1$ recover confidence-only and vision-only scoring, respectively. At $\alpha = 0.5$, $Z_{i,0.5} = \sqrt{V_i C_i}$, which is rank-equivalent to the VisConf product $V_i C_i$. The post-hoc sweep in Figure 7 shows that performance is strongest under broadly balanced weighting and varies smoothly around the symmetric choice, indicating that VisConf is not narrowly dependent on $\alpha = 0.5$.

E Analysis of Rank-Weighted Consensus

E.1 Answer Support and Rollout Quality

RWC combines answer frequency with rollout quality. Figure 8 shows that, at fixed same-answer support, higher VisConf-ranked rollouts are more accurate. Likewise, at fixed rank, greater answer support is associated with higher accuracy. This supports combining both signals in RWC.

E.2 Rescues and Regressions by Rollout Correctness Frequency

Table 7: Rescues and regressions of VisConf + RWC relative to Self-Consistency on InternVL3-8B at $N = 32$, pooled across MathVista, MMMU-Pro, and MMStar.

Correct rollouts	Questions	Rescues	Regressions	Net
1–4	521	6	1	+5
5–8	355	18	1	+17
9–12	340	52	17	+35
13–15	197	34	10	+24
1–15	**1,413**	**110**	**29**	+81
16–32	2,397	0	24	−24

To localize the paired changes underlying the gains in Figure 4, we partition questions by the number of correct rollouts in the $N = 32$ pool. A *rescue* occurs when Self-Consistency is wrong but VisConf + RWC is correct; a *regression* is the reverse. Table 7 reports these transitions on the same InternVL3-8B evaluation used in Figure 4.

E.3 Robustness Properties of Rank-Weighted Consensus

RWC depends only on the ordering of VisConf scores. Consequently, any strictly increasing transformation of the scores leaves the rollout weights and selected answer unchanged. This makes aggregation insensitive to the absolute scale of VisConf.

For $w_i = (N - r_i)/(N - 1)$, the rank weights also satisfy

$$\sum_{i=1}^{N} w_i = \frac{N}{2}, \qquad \frac{w_i}{\sum_j w_j} \leq \frac{2}{N}.$$

Thus, no individual rollout contributes more than $2/N$ of the normalized RWC mass. An isolated extreme score therefore cannot dominate the decision through its magnitude; an answer must instead accumulate support across high-ranked trajectories.

E.4 Borda and RWC Aggregation Rules

Table 8: Accuracy (%) on Qwen2.5-VL-3B, obtained by applying RWC or parameterized Borda aggregation to the same VisConf rollout scores. Avg is the unweighted mean across MathVista, MMMU-Pro, and MMStar for $N \in \{8, 16, 32\}$.

Aggregation	MathVista			MMMU-Pro			MMStar			Avg.
	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	$N=8$	$N=16$	$N=32$	
RWC	61.08	62.97	**64.50**	**46.12**	**47.24**	**48.38**	55.06	**55.75**	56.00	**55.23**
Borda ($p = 0.3$)	**61.46**	63.02	64.10	45.79	46.80	47.28	55.00	55.59	55.73	54.97
Borda ($p = 0.7$)	**61.46**	**63.11**	64.40	46.04	47.09	48.15	**55.10**	55.69	55.87	55.21
Borda ($p = 1.2$)	61.14	62.91	64.10	46.10	47.20	**48.38**	55.07	55.74	**56.07**	55.19
Borda ($p = 2.0$)	60.41	62.43	63.60	45.92	47.12	48.32	54.85	55.73	56.00	54.93

Parameterized Borda aggregation assigns

$$b_p(r_i) = (N - r_i + 1)^p.$$

For $p = 1$, this is an affine transformation of the RWC rollout weight:

$$b_1(r_i) = (N - 1)w_i + 1.$$

Consequently, its answer-group score satisfies

$$S_{p=1}(a) = (N - 1)S_{\mathrm{RWC}}(a) + n_a,$$

Therefore, linear Borda adds an answer-frequency term to the rank-weighted quality evidence. This distinction is relevant to the failure mode studied in Section 7.3 when correct trajectories are high-quality but remain a minority: the additive n_a term in linear Borda explicitly favors the more frequent answer, partially restoring the majority bias that RWC is intended to overcome. More generally, $p < 1$ further compresses rank differences and approaches unweighted consensus as $p \to 0$, whereas $p > 1$ increasingly concentrates influence on the highest-ranked trajectories. RWC instead uses a fixed linear rank schedule with no additive frequency bonus.

Across the range of evaluated p values, Borda aggregation is broadly comparable to RWC, reinforcing that rank-weighted answer aggregation is generally effective for combining rollout quality with consensus. RWC offers a simpler alternative by fixing the weighting schedule and removing the need to tune the exponent p, while retaining comparable performance across models, datasets, and sampling budgets.

F Qualitative Examples

We present five examples at $N = 32$ to illustrate distinct qualitative outcomes: two corrections by VisConf + RWC, one regression, one case in which all methods succeed, and one case in which all methods fail. The examples span two models and all three benchmarks. For each example, *support* denotes either the number of votes for the selected answer under Self-Consistency or the sum of normalized within-question rank weights under RWC. LookBack top-1 selects a single rollout and therefore has no answer-group support value.

F.1 Corrections by VisConf + RWC

(a) Visual proportion estimation – Qwen2.5-VL-3B, MMStar sample 359



Question. Roughly how much of the image is occupied by the two people on the bench? The relevant options are A: more than 50% and B: less than 30%.

Gold answer: B (less than 30%)

Candidate pool. A receives 16 rollouts, B receives 15, and D receives one.

Method	Prediction	Outcome	Support
Self-Consistency	A	incorrect	16 votes
LookBack top-1	A	incorrect	–
LookBack + RWC	A	incorrect	8.48
VisConf + RWC	**B**	correct	**9.06**

Interpretation. The unweighted vote is wrong by one rollout. Correct trajectories explicitly compare the people with the much larger water and sky regions, whereas many A trajectories treat the people as the image's semantic focus and infer that they must occupy most of its area. VisConf + RWC gives the minority B group enough collective weight to reverse this narrow incorrect plurality.

(b) Tacheometric calculation – InternVL3-8B, MMMU-Pro sample
`test_Architecture_and_Engineering_492`

Instrument station	*Staff station*	*Vertical angle*	*Staff readings (m)*	*Remarks (m)*
	B.M.	– 6°0′	1.100, 1.153, 2.060	R.L. of B.M.
A	*B*	+ 8°0′	0.982, 1.105, 1.188	= 976.00 m

Question. Using the staff readings in the figure, determine the horizontal distance from A to B and the reduced level of B.

Gold answer: D: $AB = 94.14\,\text{m}$ and R.L. $= 988.81\,\text{m}$

Candidate pool. C receives 14 rollouts, D receives 13, A and no-answer receive two each, and B receives one.

Method	Prediction	Outcome	Support
Self-Consistency	C	incorrect	14 votes
LookBack top-1	C	incorrect	–
LookBack + RWC	C	incorrect	7.03
VisConf + RWC	**D**	correct	**7.61**

Interpretation. The candidate pool is fragmented and the correct answer trails the leading incorrect answer by one rollout. The incorrect C trajectories typically reach a nearby horizontal-distance value but propagate an error through the trigonometric or reduced-level calculation. VisConf + RWC shifts the decision to the D group, whose trajectories more consistently retain the required quantities through the complete calculation.

F.2 A Regression

(c) Relational counting – InternVL3-8B, MathVista sample 984



Question. Is the number of gray cars less than the number of small metallic minivans? The options are A: yes and B: no.

Gold answer: B (no)

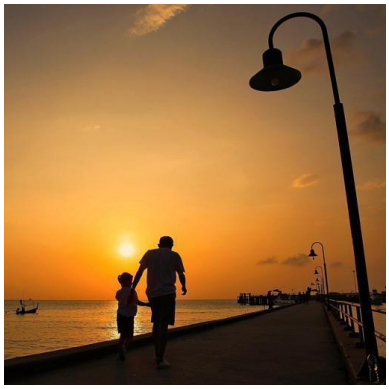
Candidate pool. B receives 17 rollouts and A receives 15.

Method	Prediction	Outcome	Support
Self-Consistency	**B**	correct	17 votes
LookBack top-1	**B**	correct	–
LookBack + RWC	**B**	correct	**8.58**
VisConf + RWC	A	incorrect	9.58

Interpretation. Here, the calibrated score is misleading. The candidate pool has a narrow but correct majority for B, and both LookBack variants preserve that decision. VisConf instead ranks enough of the incorrect A trajectories highly to reverse the result. This example illustrates that confidence and visual engagement remain imperfect quality signals: strong engagement with an image does not guarantee that the depicted objects were counted or categorized correctly.

F.3 Agreement and Shared Failure

(d) Mood recognition – Qwen2.5-VL-3B, MMStar sample 13



Question. Which mood does the image convey? The relevant options are B: cozy and C: happy.

Gold answer: B (cozy)

Candidate pool. B receives 16 rollouts, C receives 15, and one rollout produces no answer.

Method	Prediction	Outcome	Support
Self-Consistency	**B**	correct	16 votes
LookBack top-1	**B**	correct	–
LookBack + RWC	**B**	correct	**9.61**
VisConf + RWC	**B**	correct	**8.23**

Interpretation. This is a closely contested but stable agreement case. Both leading groups recognize the warm sunset and the companionship between the adult and child; they differ mainly in whether those cues imply "cozy" or "happy." Despite the one-vote margin, both rollout-quality scores and both aggregation rules retain the correct B decision.

(e) Dominant-color recognition – Qwen2.5-VL-3B, MMStar sample 276



Question. What is the dominant color in the image? The relevant options are B: brown and C: black.

Gold answer: B (brown)

Candidate pool. C receives 17 rollouts and B receives 15. Thus, the pool contains a substantial minority of correct trajectories.

Method	Prediction	Outcome	Support
Self-Consistency	C	incorrect	17 votes
LookBack top-1	C	incorrect	–
LookBack + RWC	C	incorrect	9.94
VisConf + RWC	C	incorrect	8.87

Interpretation. This is a shared failure despite the presence of substantial correct evidence. The C trajectories emphasize individually black objects, such as clothing, instruments, and acoustic panels, while the B trajectories attend to the image-wide brown or sepia cast. Both quality scores reinforce the narrow incorrect majority rather than recovering the correct minority.